Introduction to Language Models: Auto-Regressive decoders, prompting, diffusion for text generation

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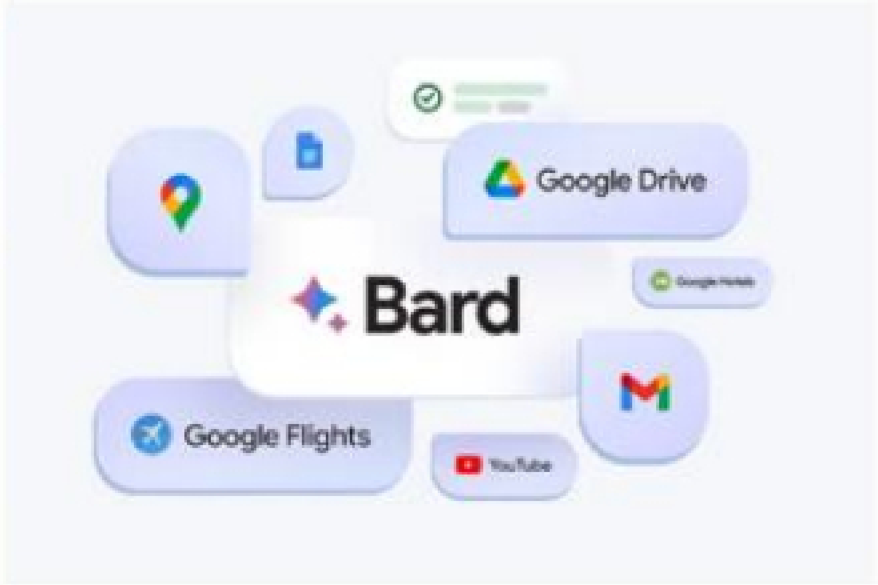
What is a Language Model (LM)?

Video from https://wayve.ai/thinking/lingo-natural-language-autonomous-driving/

What is a Language Model (LM)?



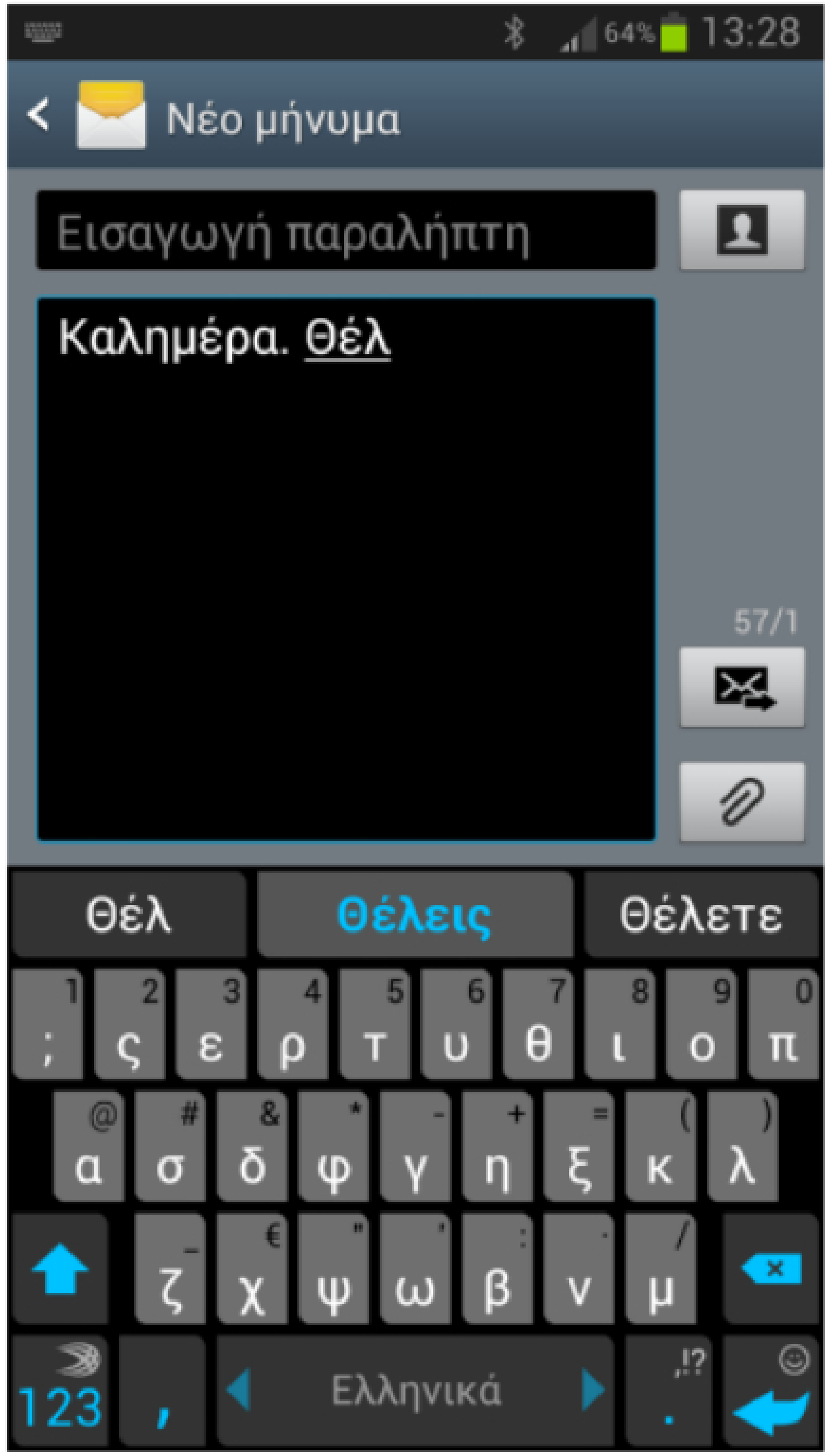

Μάθημα-πείραμα με το ChatGPT στο Πανεπιστήμιο Αθηνών

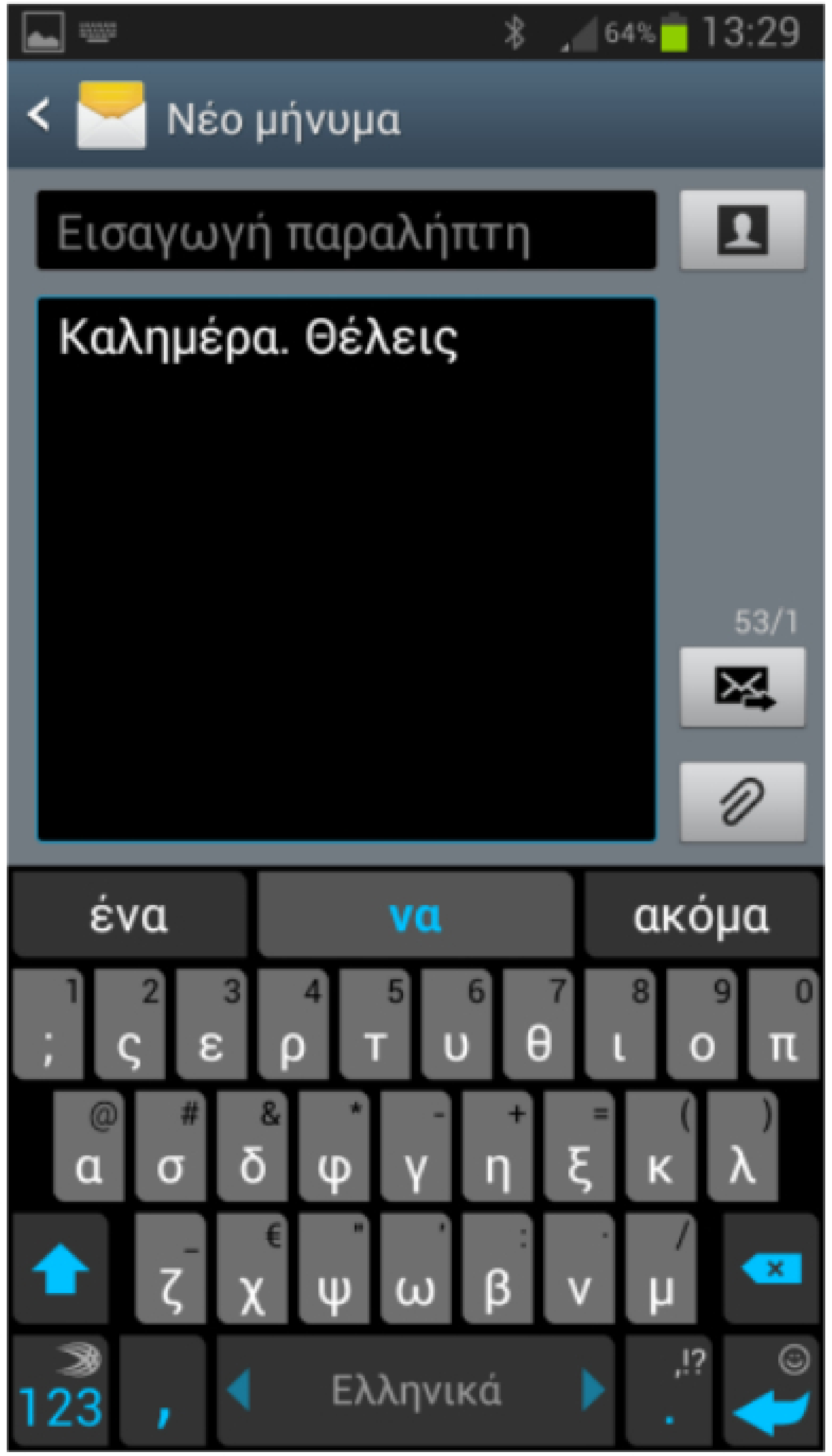
Αναλύοντας τη λειτουργία του ChatGPT

Η βάση της Τεχνητής Νοημοσύνης είναι η μηχανική μάθηση, στο πλαίσιο της οποίας δίνονται στον υπολογιστή πολλά «παραδείγματα εκπαίδευσης» και υποδεικνύονται οι επιθυμητές λειτουργίες/απόκαι. «Ο υπολογιστής "μαθαίνει" με αλγόριθμους μηχανικής μάθησης, οι οποίοι του επιτρέπουν να μαθαίνει μόνος του από παραδείγματα ώστε να εκτελεί, στη συνέχεια, μία εργασία σχετικά πολύπλοκη. Η αναγνώριση ομιλίας ή η αυτόνομη οδήγηση των οχημάτων είναι μερικά μόνο παραδείγματα μηχανικής μάθησης με παραδείγματα εκπαίδευσης.

Τα τελευταία δέκα χρόνια έχει κυριαρχήσει ένα παρακλάδι της μηχανικής μάθησης: η "βαθιά μάθηση" με εντυπωσιακά αποτελέσματα, όπως το ChatGP», εξηγεί ο δρ Ίων Ανδρουτσόπουλος, καθηγητής Τεχνητής Νοημοσύνης στο Τμήμα Πληροφορικής του Οικονομικού Πανεπιστημίου Αθηνών. Προκειμένου να γίνει κατανοητή η «βαθιά μάθηση», αναφέρει: «Πρόκειται για ένα είδος μηχανικής μάθησης, που βασίζεται σε περίπλοκα, μεγάλης κλίμακας νευρωνικά δίκτυα. Κάθε δίκτυο αποτελείται από πολύ μικρές μονάδες επεξεργασίας, που συνδέονται μεταξύ τους και τις ονομάζουμε νευρώνες. Ο κάθε ένας από αυτούς παίρνει είσοδο από πολλούς άλλους νευρώνες και ταυτόχρονα δίνει την έξοδό του επίσης σε πολλούς νευρώνες.

What is a Language Model (LM)?



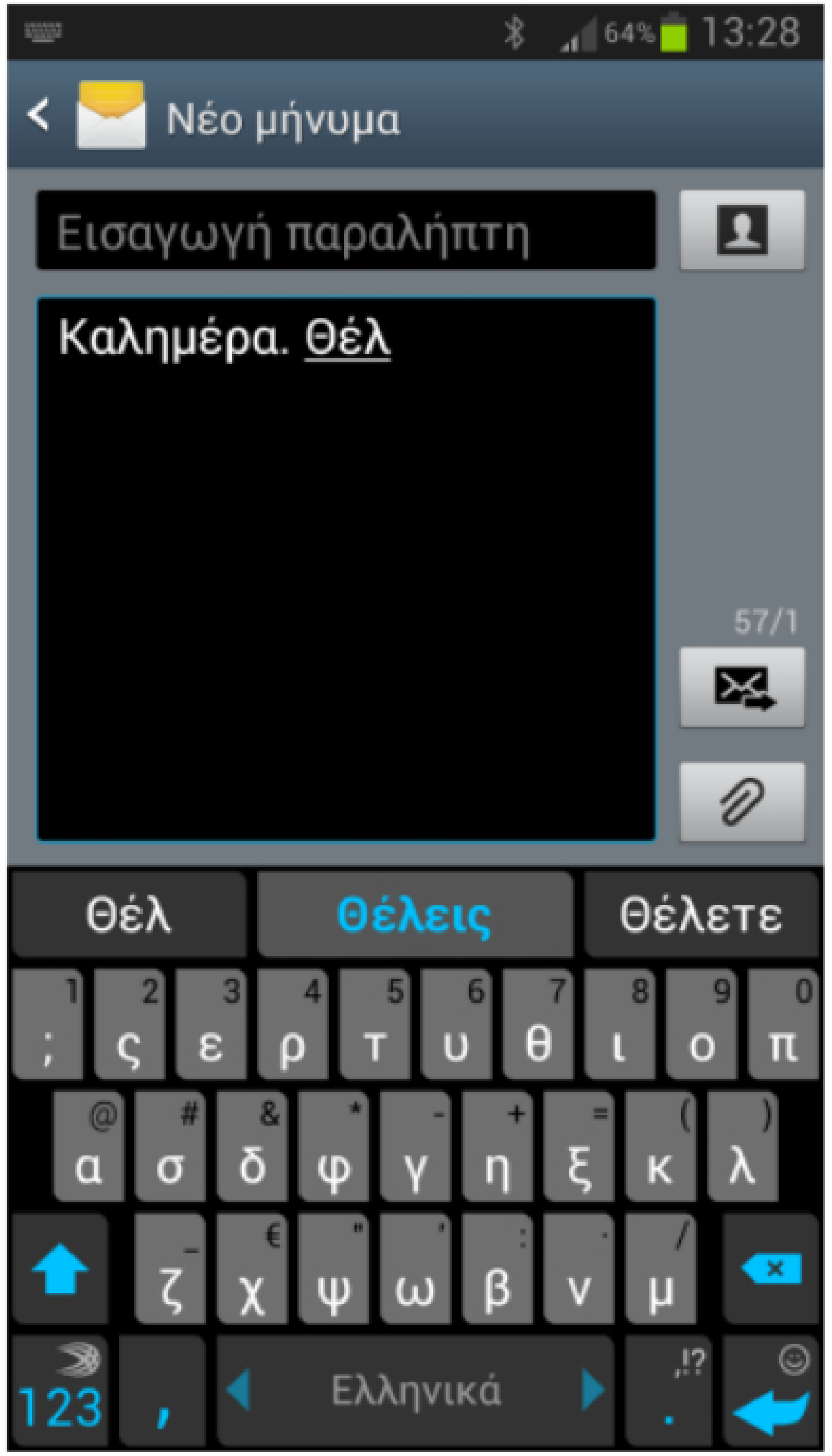


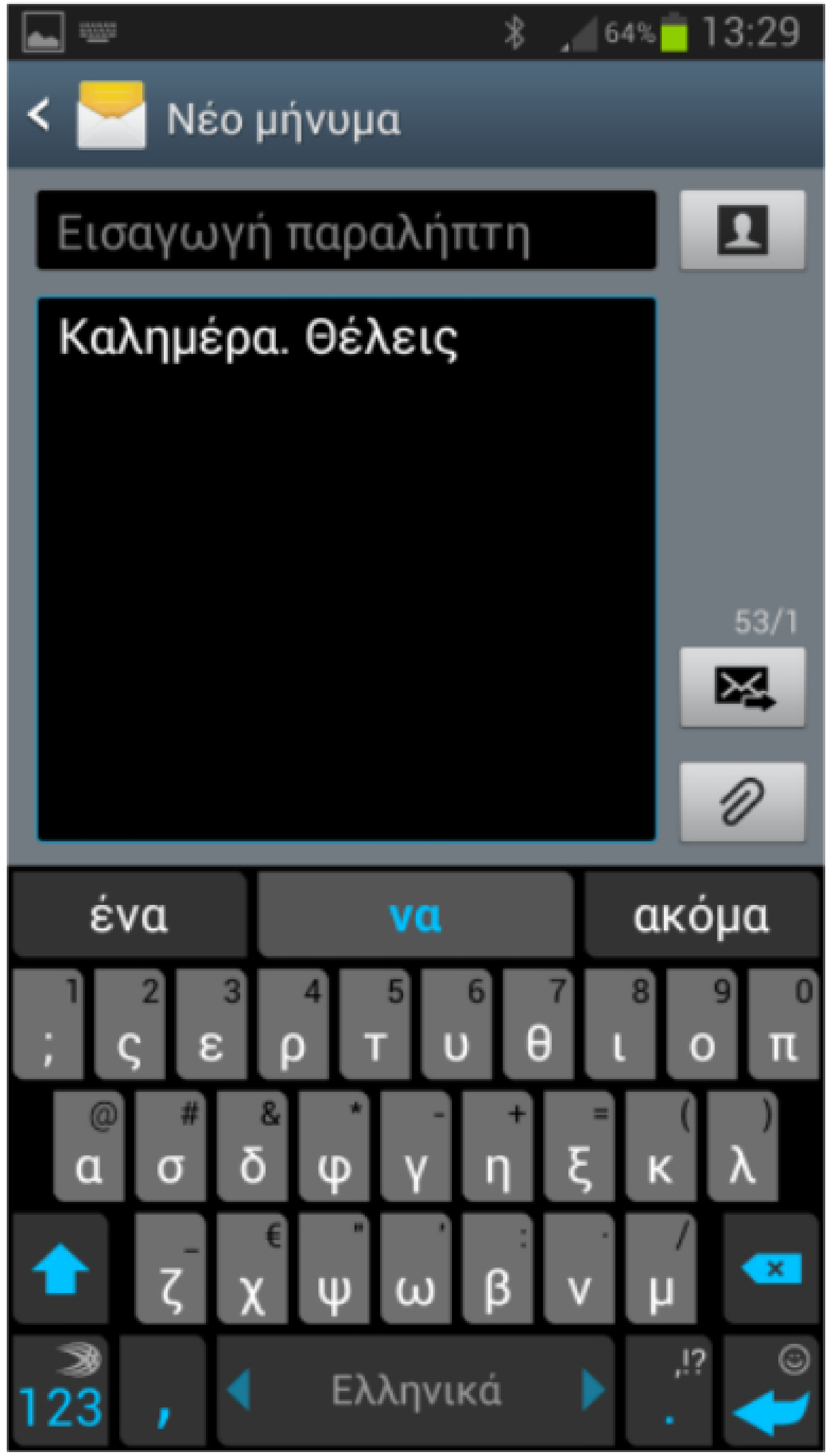
$$\overrightarrow{\text{καλημέρα}} = \begin{bmatrix}1.8\\2.3\\-1.4\\3.7\\\dots\\-1.1\end{bmatrix} \qquad \overrightarrow{\text{θέλεις}} = \begin{bmatrix}2.4\\-3\\9.3\\5.1\\\dots\\3.9\end{bmatrix}$$

$$\overrightarrow{\text{πορτοκαλάδα}} = \begin{bmatrix}2.2\\3.8\\1.2\\-6.4\\\dots\\7.1\end{bmatrix}$$

$$P(\overrightarrow{\text{πορτοκαλάδα}} \mid \overrightarrow{\text{καλημέρα}}\ \overrightarrow{\text{θέλεις}}) = ?$$

What is a Language Model (LM)?





$$P\left(\overrightarrow{\text{πορτοκαλάδα}} \mid \overrightarrow{\text{καλημέρα}}\ \overrightarrow{\text{θέλεις}}\right)$$

$$\overrightarrow{\text{πορτοκαλάδα}} = \begin{bmatrix} 2.2 \\ 3.8 \\ 1.2 \\ -6.4 \\ \dots \\ 7.1 \end{bmatrix}$$

98%

$\overrightarrow{\text{βυσσινάδα}} \longrightarrow$ **95%**

$\overrightarrow{\text{καφέ}} \longrightarrow$ **97%**

$\vdots$ $\vdots$

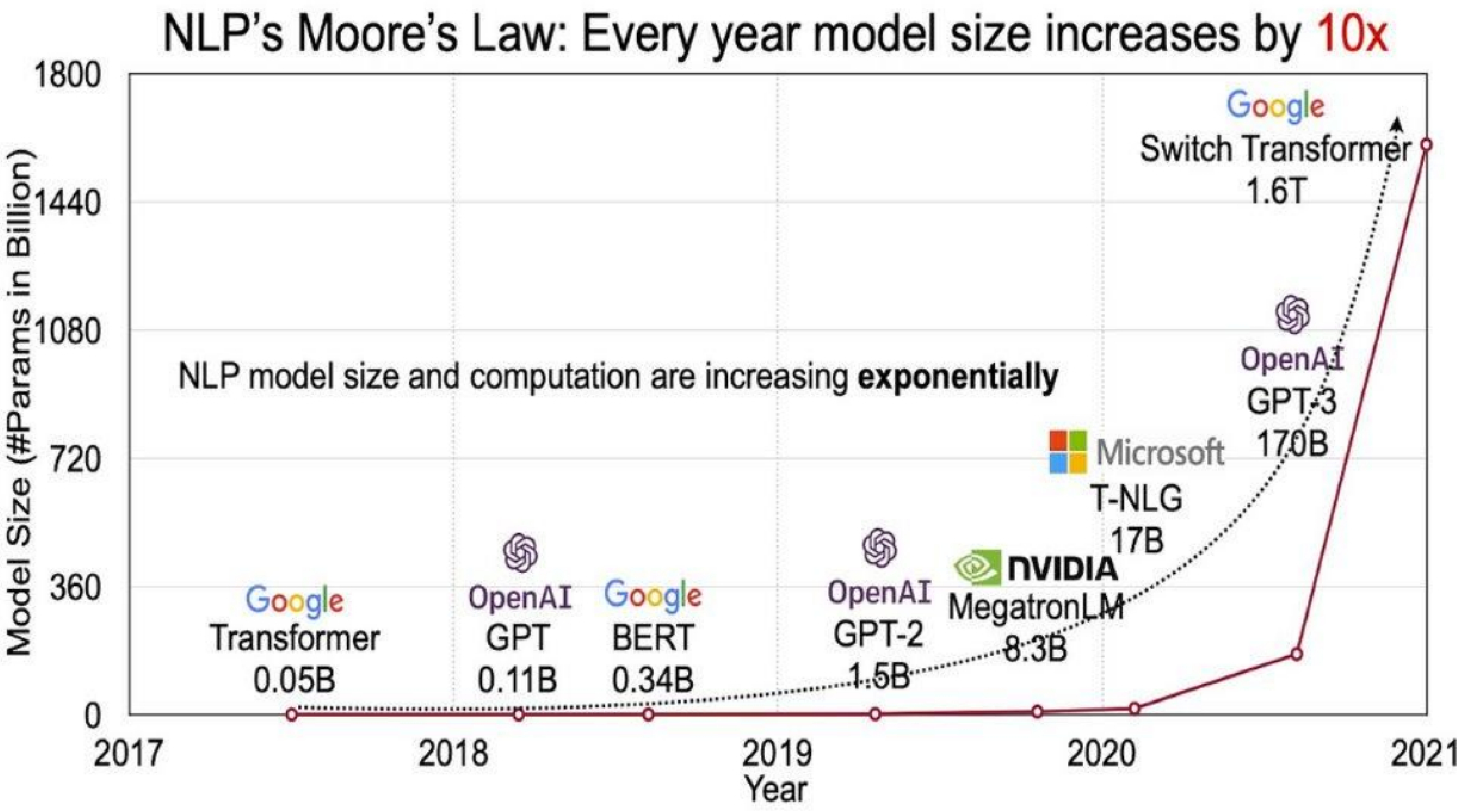
$\overrightarrow{\text{αμετροεπής}} \longrightarrow$ **2%**

$\overrightarrow{\text{διάτρητος}} \longrightarrow$ **1%**

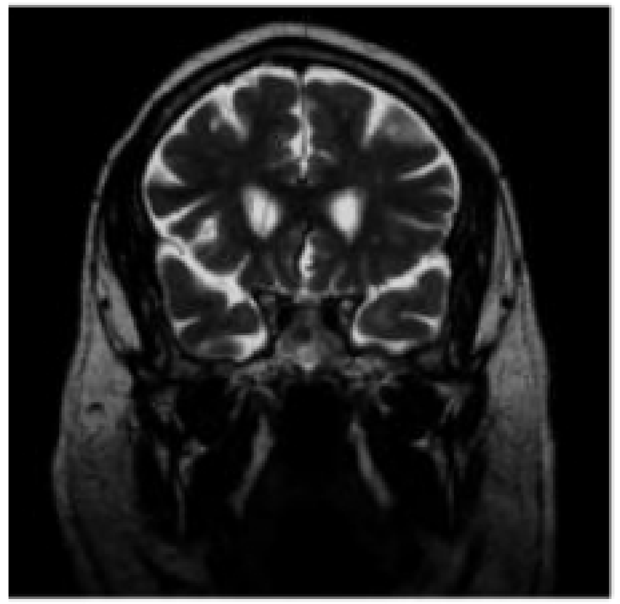
Known Language Models: OpenAI GPT

Known Language Models: Meta LlaMA

Why do Large LMs perform good?



...in different domains?



You

You are an expert in medical imaging. For the provided medical image, you must answer with a caption of the medical abnormalities it depicts if there are any.

ChatGPT

The image provided appears to be an MRI scan of the human brain. In the image, we can see a clear indication of a ventricular system with what seems like an abnormal enlargement of the ventricles, a condition known as hydrocephalus. Moreover, there is a line with a measurement annotation which could suggest that a radiologist or a medical professional was measuring the size of the ventricles to quantify the extent of ventricular enlargement.

Hydrocephalus is a condition where there is an accumulation of cerebrospinal fluid (CSF) within the ventricles, which can lead to increased intracranial pressure. It can be due to an obstruction of CSF flow, poor absorption, or excessive production. The treatment often involves the surgical insertion of a shunt system to drain the excess fluid, or an endoscopic third ventriculostomy to allow the fluid to bypass the obstruction.

It's important to note that a definitive diagnosis should only be made by a qualified medical professional with access to the patient's history, physical examination, and full imaging studies. The provided information is based solely on the visible features in the image and does not constitute a medical diagnosis.

Is the text meaningful or purely hallucinated?

Prompting Language Models

- **In-context learning:**
  A frozen LM performs a task only by conditioning on the prompt text.
- **Few-shot in-context learning:**
  - In the case of **prompting** the following apply:
    - The prompt includes examples of the intended behavior,
    - no examples of the intended behavior were seen in training.
  - We are unlikely to be able to verify.
  - In the case of **supervised learning**, it has the sense of "training on few examples". The above is different.
- **Zero-shot in-context learning:**
  - In the case of **prompting** the following apply:
    - The prompt includes no examples of the intended behavior
    - However, the prompt can contain other instructions, as in InstructGPT or ChatGPT,
    - no examples of the intended behavior were seen in training.
  - We are unlikely to be able to verify.
  - Formatting and other instructions seem like a gray area, but we will allow them in this category.

Chain-of-Thought (CoT) Prompting

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27.

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had 23 - 20 = 3. They bought 6 more apples, so they have 3 + 6 = 9. The answer is 9.

Manually annotated chain-of-thought to a pretrained LLM.

Automatically written chain-of-thought is by the same LLM.

Figure 1 shows an example of a model producing a chain of thought to solve a math word problem that it would have otherwise gotten incorrect. The chain of thought in this case resembles a solution and can interpreted as one, but we still opt to call it a chain of thought to better capture the idea that it mimics a step-by-step thought process for arriving at the answer (and also, solutions/explanations typically come *after* the final answer (Narang et al., 2020; Wiegreffe et al., 2022; Lampinen et al., 2022, *inter alia*)).

There is also zero-shot Chain-of-Thought that just **adds a preamble** to the prompt…

CoT with LM-Designed prompts

No.	Category	Zero-shot CoT Trigger Prompt	Accuracy
1	LM-Designed	Let's work this out in a step by step way to be sure we have the right answer.	**82.0**
2	Human-Designed	Let's think step by step. (*1)	78.7
3		First, (*2)	77.3
4		Let's think about this logically.	74.5
5		Let's solve this problem by splitting it into steps. (*3)	72.2
6		Let's be realistic and think step by step.	70.8
7		Let's think like a detective step by step.	70.3
8		Let's think	57.5
9		Before we dive into the answer,	55.7
10		The answer is after the proof.	45.7
-		(Zero-shot)	17.7

Using a **learnable prompt** improves the LM's downstream performance!

Stepwise zero-shot CoT

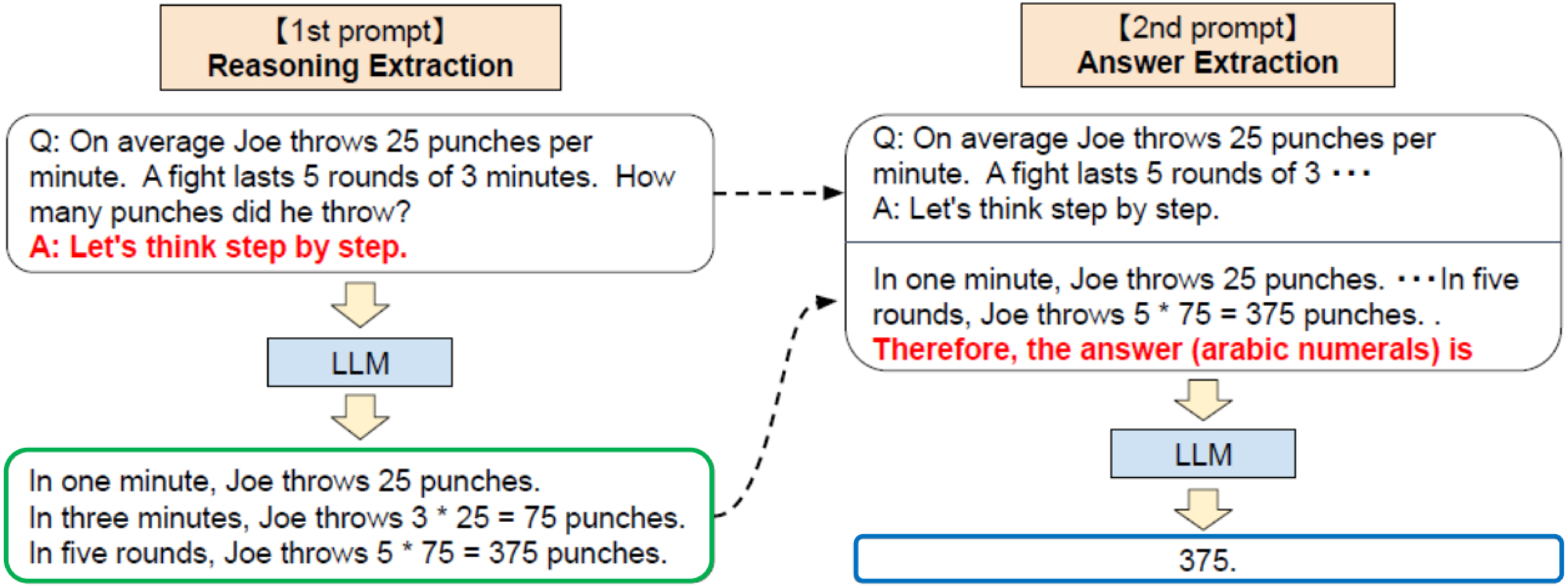
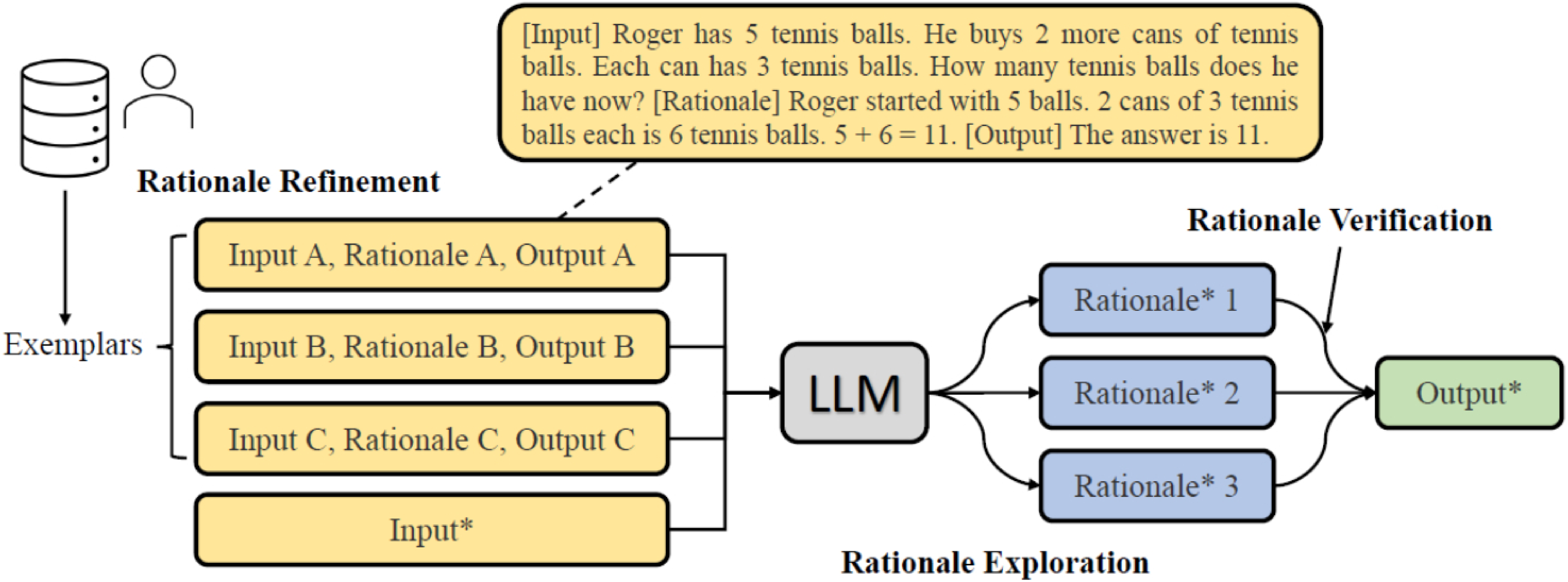


Figure 2: Full pipeline of Zero-shot-CoT as described in § 3: we first use the first "reasoning" prompt to extract a full reasoning path from a language model, and then use the second "answer" prompt to extract the answer in the correct format from the reasoning text.

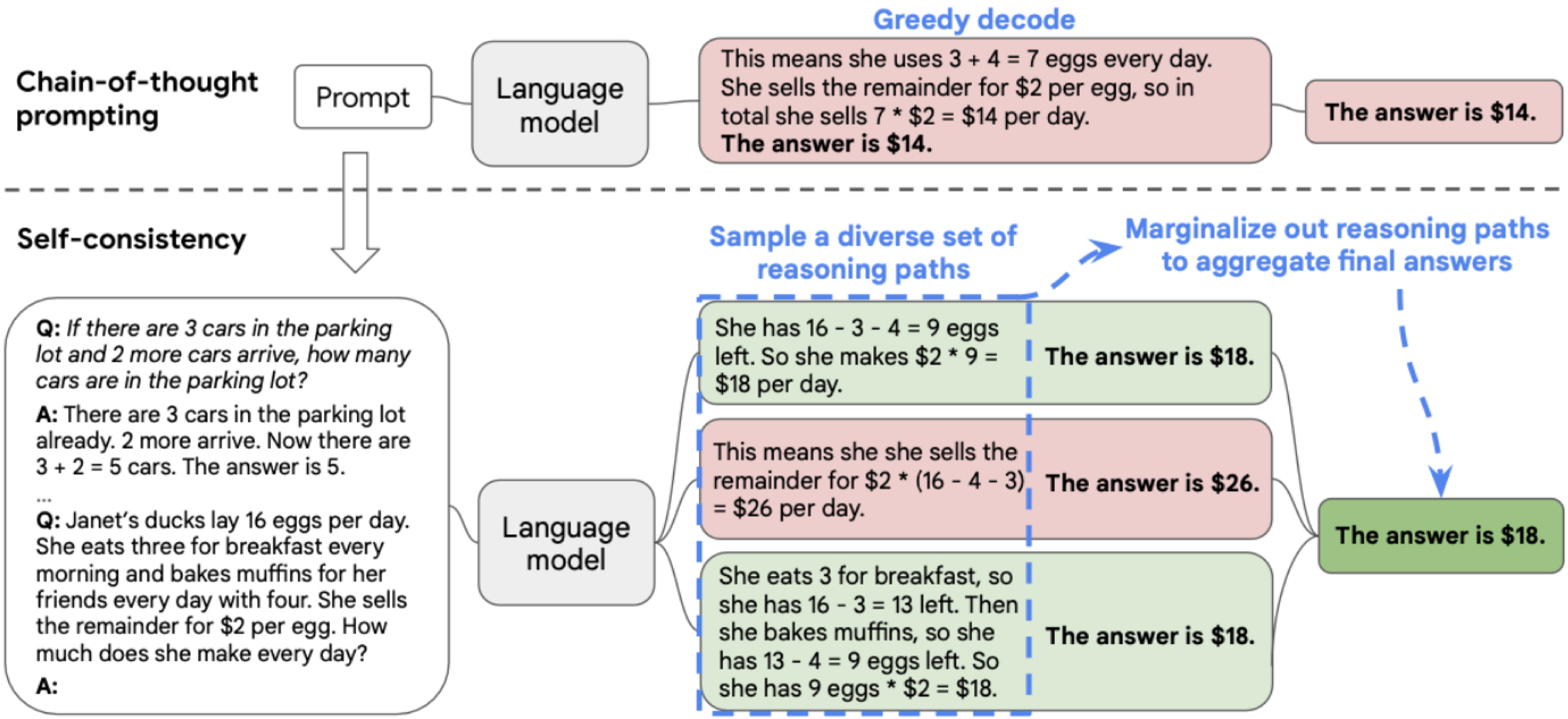
Using **multiple prompts** improves the zero-shot performance of CoT!

Rationale ensembling strategies



- **Rationale refinement:**
  - Creating more effective examples of reasoning steps, to better elicit reasoning in LLMs.
  - Complexity-based prompting, algorithmic prompting, AutoCoT etc.
- **Rationale exploration and verification:**
  They involve exploring and verifying the rationales produced by LLMs.

Rationale exploration: self-consistency



The diverse set of rationales permits us to choose the most confident answer!

Rationale-Augmented Ensembles

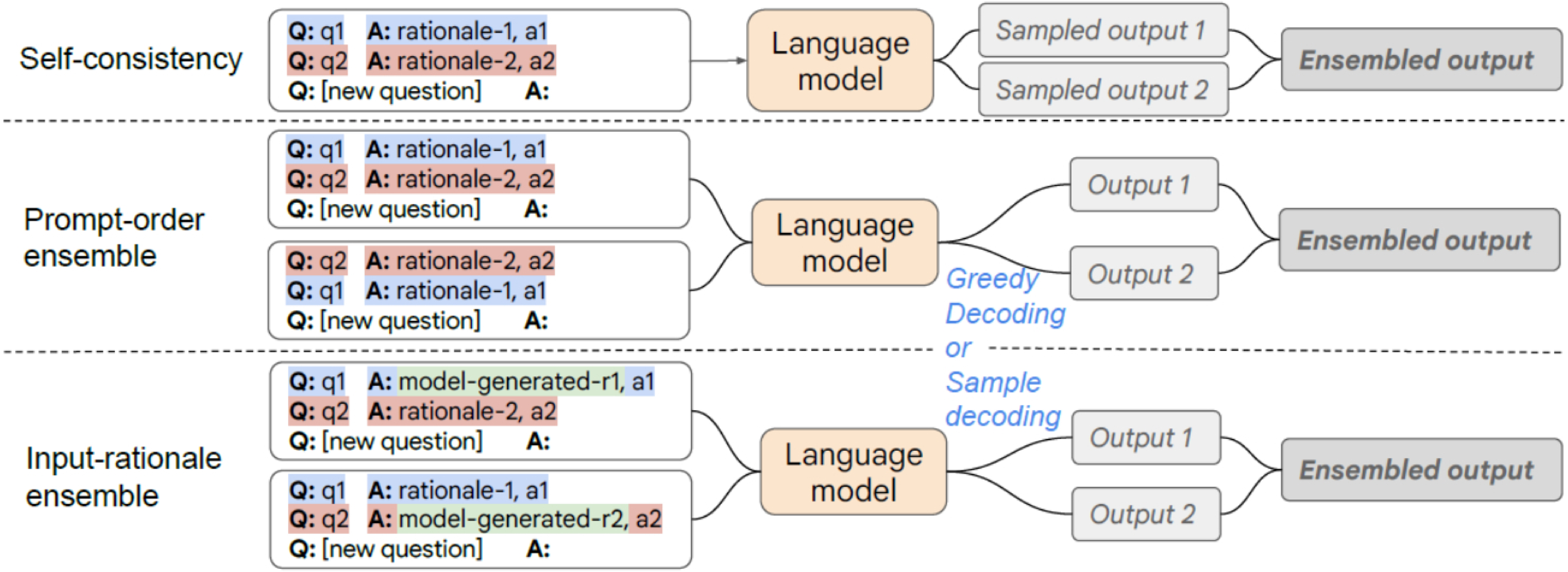


Figure 1: An overview of different ways of composing rationale-augmented ensembles, depending on how the randomness of rationales is introduced. Here q, a, r correspond to question, answer, and rationale, respectively. Rationales are human-written unless specified as model-generated.

RAG: Retrieval Augmented Generation

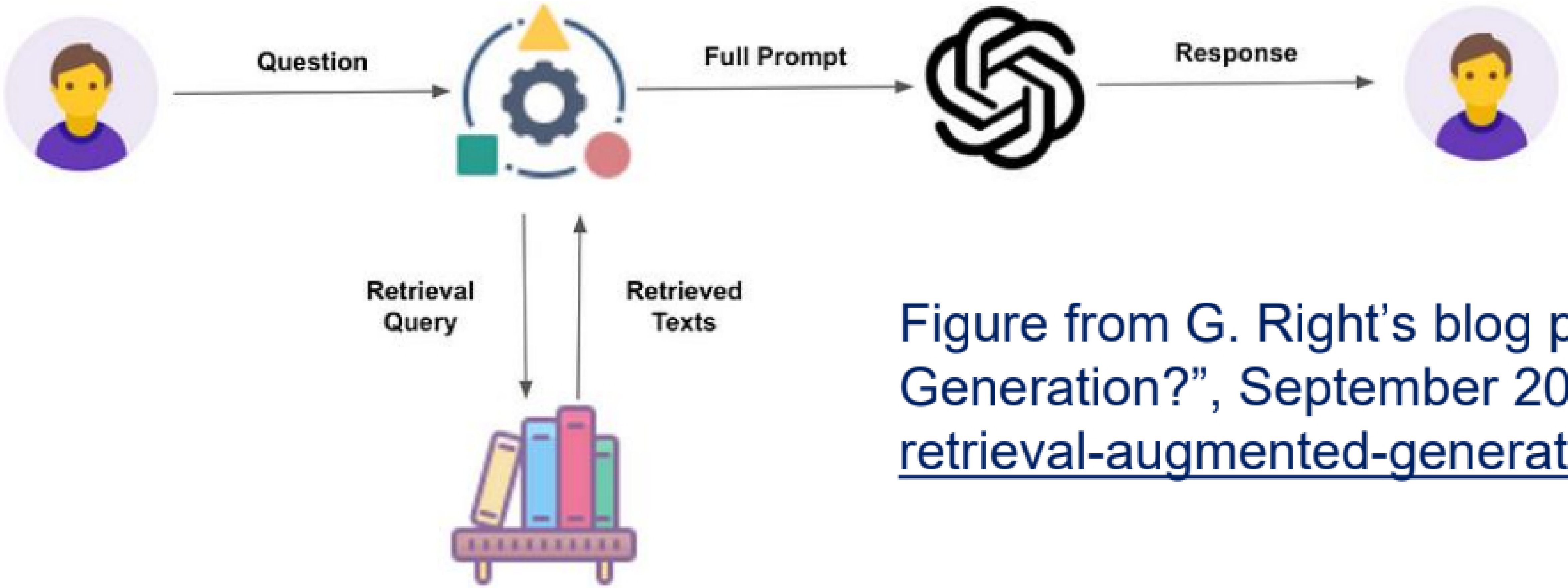


Figure from G. Right's blog post, "What is Retrieval Augmented Generation?", September 2023 (https://www.linkedin.com/pulse/what-retrieval-augmented-generation-grow-right/).

- Learns to jointly retrieve and generate new text end-to-end.
- Does not need labels for the relevant documents (in principle).
- "Frozen RAG" can be used for any seq-to-seq downstream task and with any LLM.
  - Black-box pretrained models put together in a pipeline...
  - We can use conventional IR (e.g., TF-IDF, BM25: DrQA) or dense retrieval (documents and questions encoded, compared via a similarity function, e.g. OrQA).
  - Input (prompt) to the LLM: question, retrieved documents (or snippets), instructions telling the LLM to base its answer on the retrieved documents.
- Works because of **in-context learning.**

Choosing the next word...

At each timestep t: $\arg\max_w P(w \mid w_{1:t-1})$

$P(\text{"nice"}\mid\text{"The"}) = 0.5$

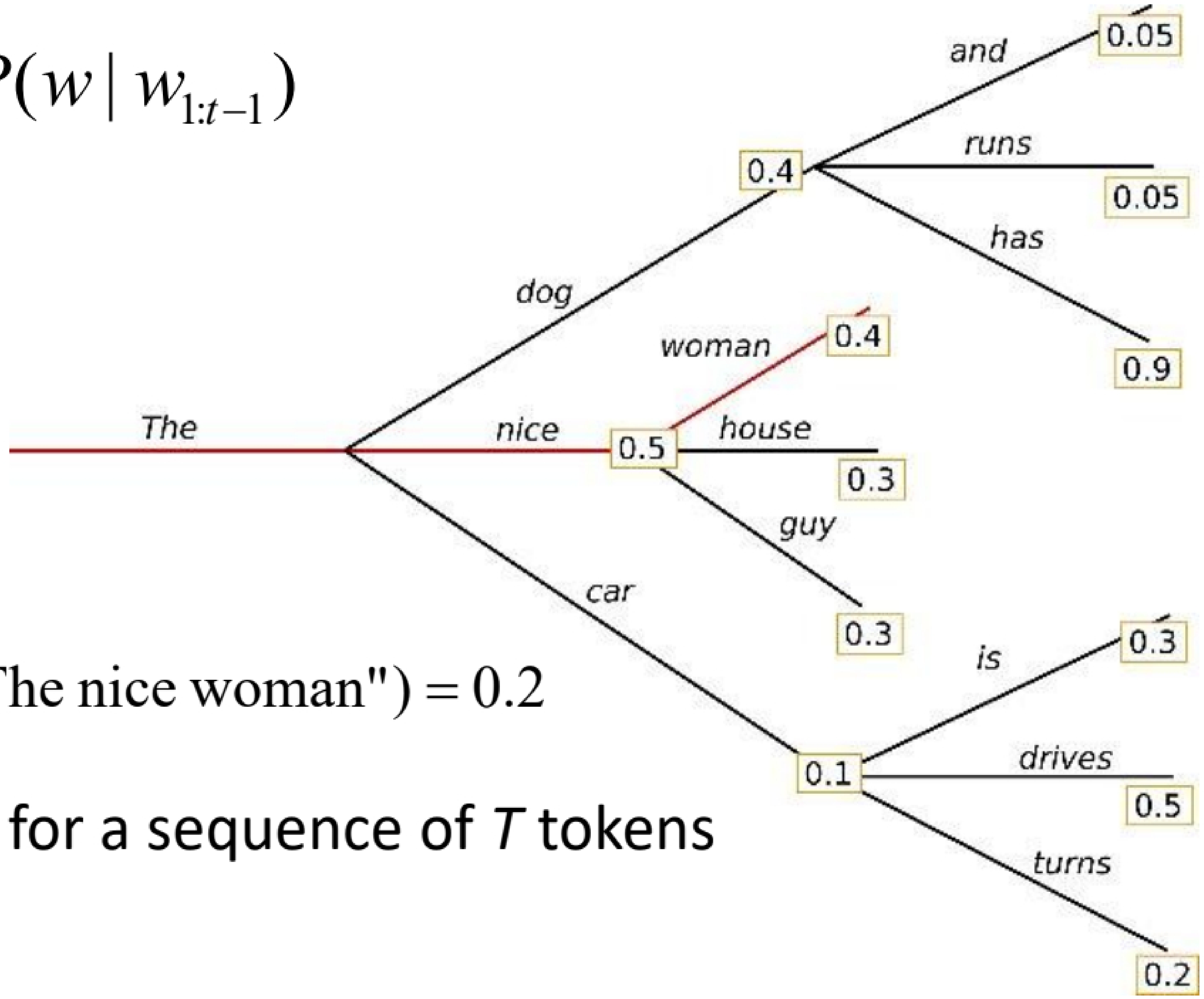
$P(\text{"dog"}\mid\text{"The"}) = 0.4$

$P(\text{"car"}\mid\text{"The"}) = 0.1$

$P(\text{"woman"}\mid\text{"The nice"}) = 0.4$

$P(\text{"house"}\mid\text{"The nice"}) = 0.3$

$P(\text{"guy"}\mid\text{"The nice"}) = 0.3$ $\quad \Rightarrow P(\text{"The nice woman"}) = 0.2$



$P(w_{1:T} \mid W_0) = \prod_{t=1}^{T} P(w_t \mid w_{1:t-1}, W_0)$ for a sequence of T tokens

$w_{1:0} = \emptyset$

Choosing the right word?

At each timestep *t*: $\arg\max_w P(w \mid w_{1:t-1})$

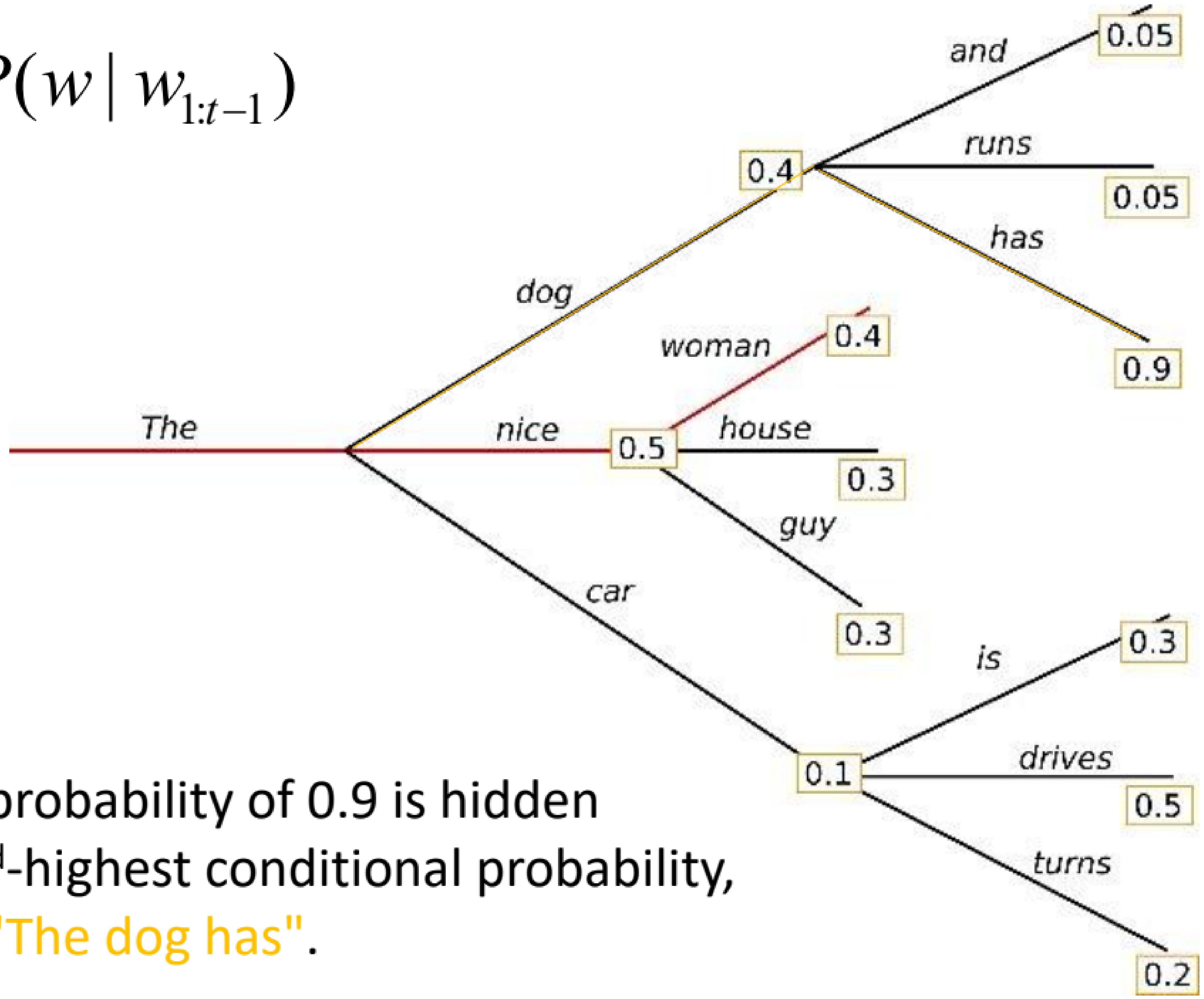
$P(\text{"nice"} \mid \text{"The"}) = 0.5$

$P(\text{"dog"} \mid \text{"The"}) = 0.4$

$P(\text{"car"} \mid \text{"The"}) = 0.1$

$P(\text{"The nice woman"}) = 0.2$

$P(\text{"The dog has"}) = 0.36$



The word "has" with its high conditional probability of 0.9 is hidden behind the word "dog", which has the 2nd-highest conditional probability, so that greedy search misses the phrase "The dog has".

Choosing the right word?

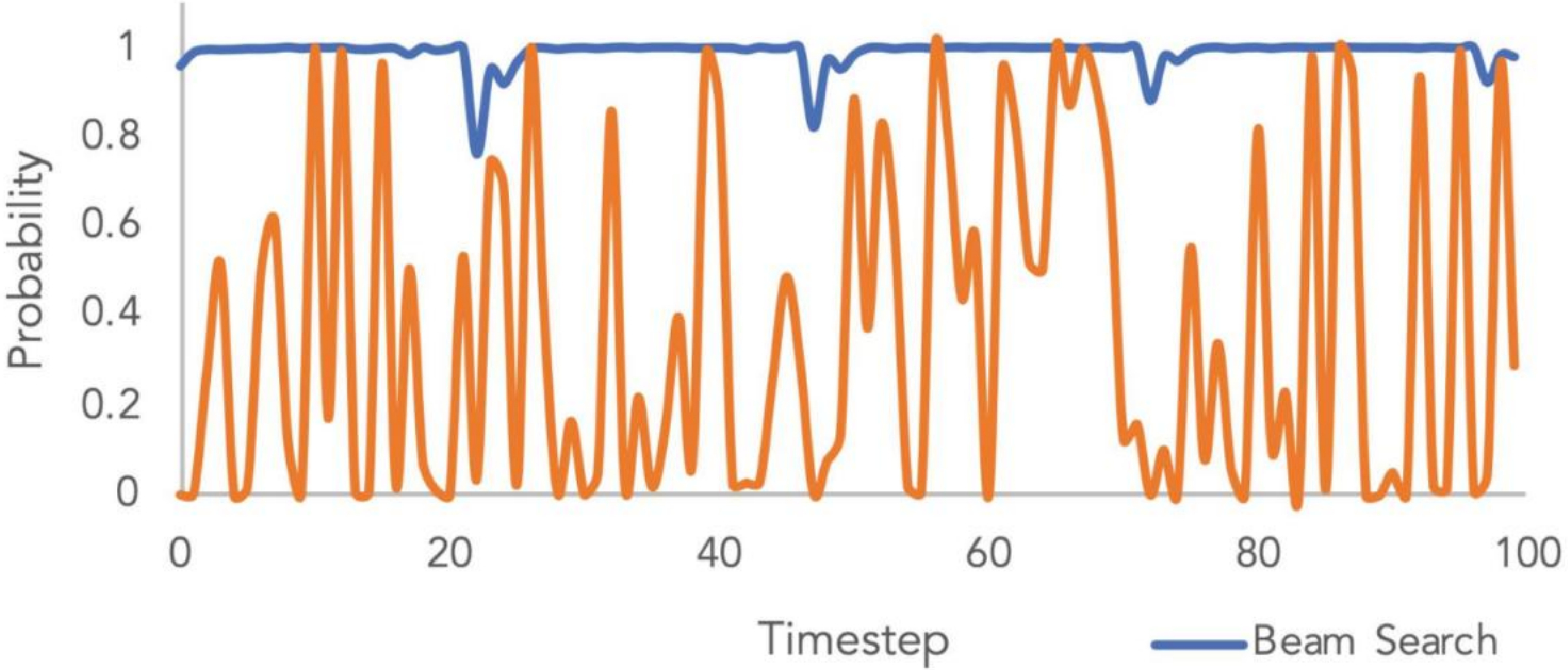
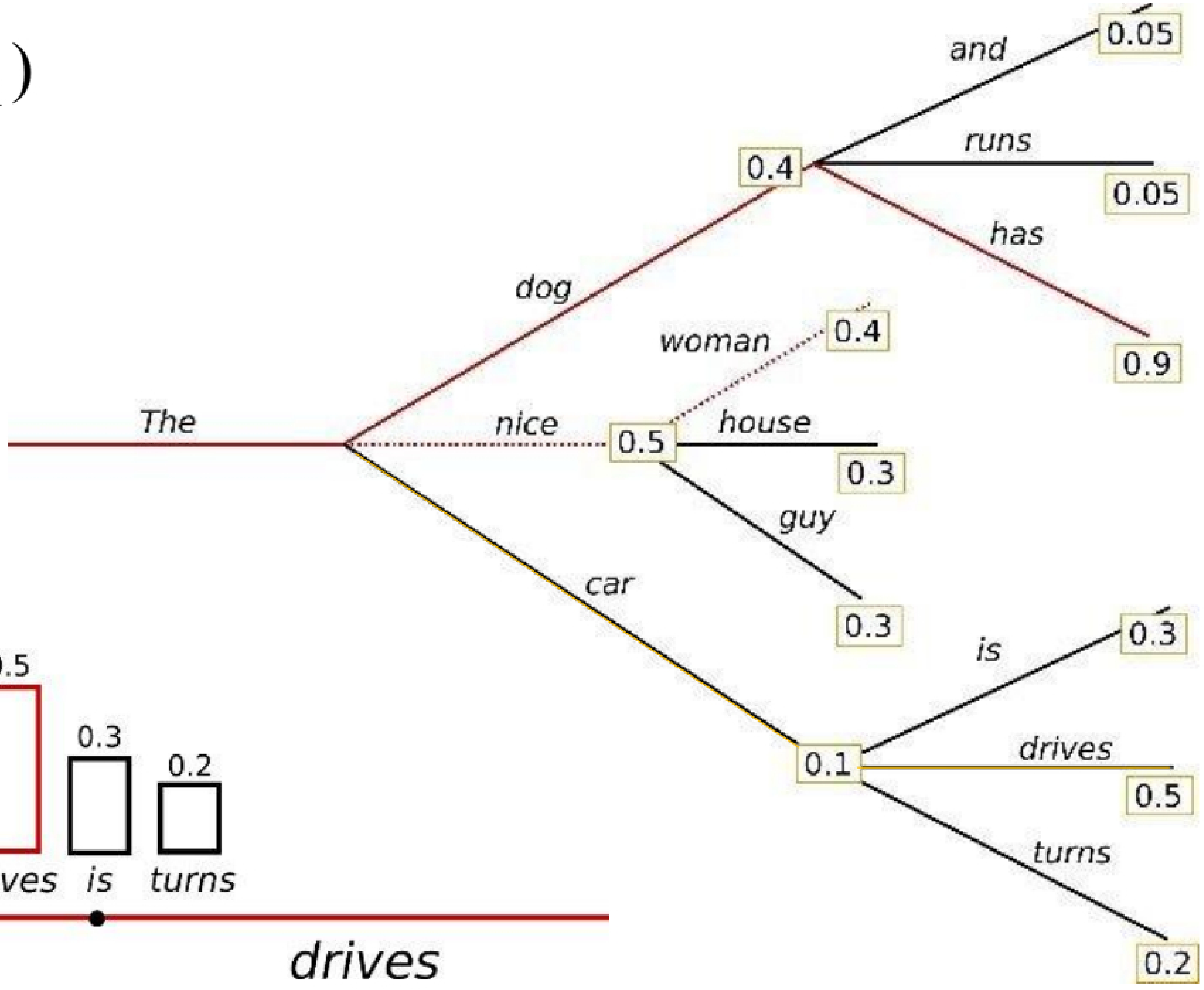


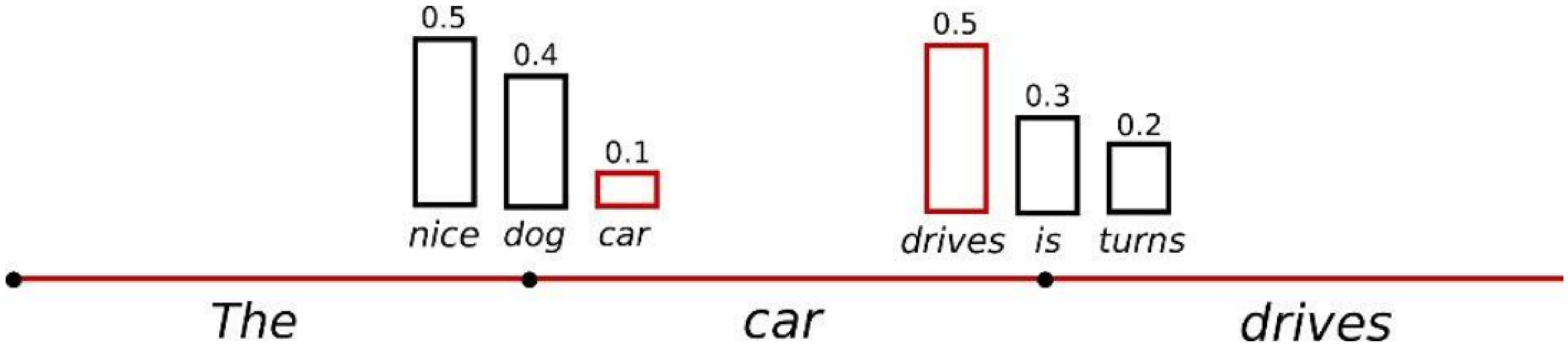
Figure from https://arxiv.org/abs/1904.09751

Sampling the next word (sometimes...)

At some timestep *t*: $w_t \sim P(w \mid w_{1:t-1})$

The material presented about sampling originates from the following blog article: https://huggingface.co/blog/how-to-generate
If interested in **advanced sampling methods** (e.g., top-*p* sampling, Nucleus) please read the full article offline.





Diffusion in images (last lecture)

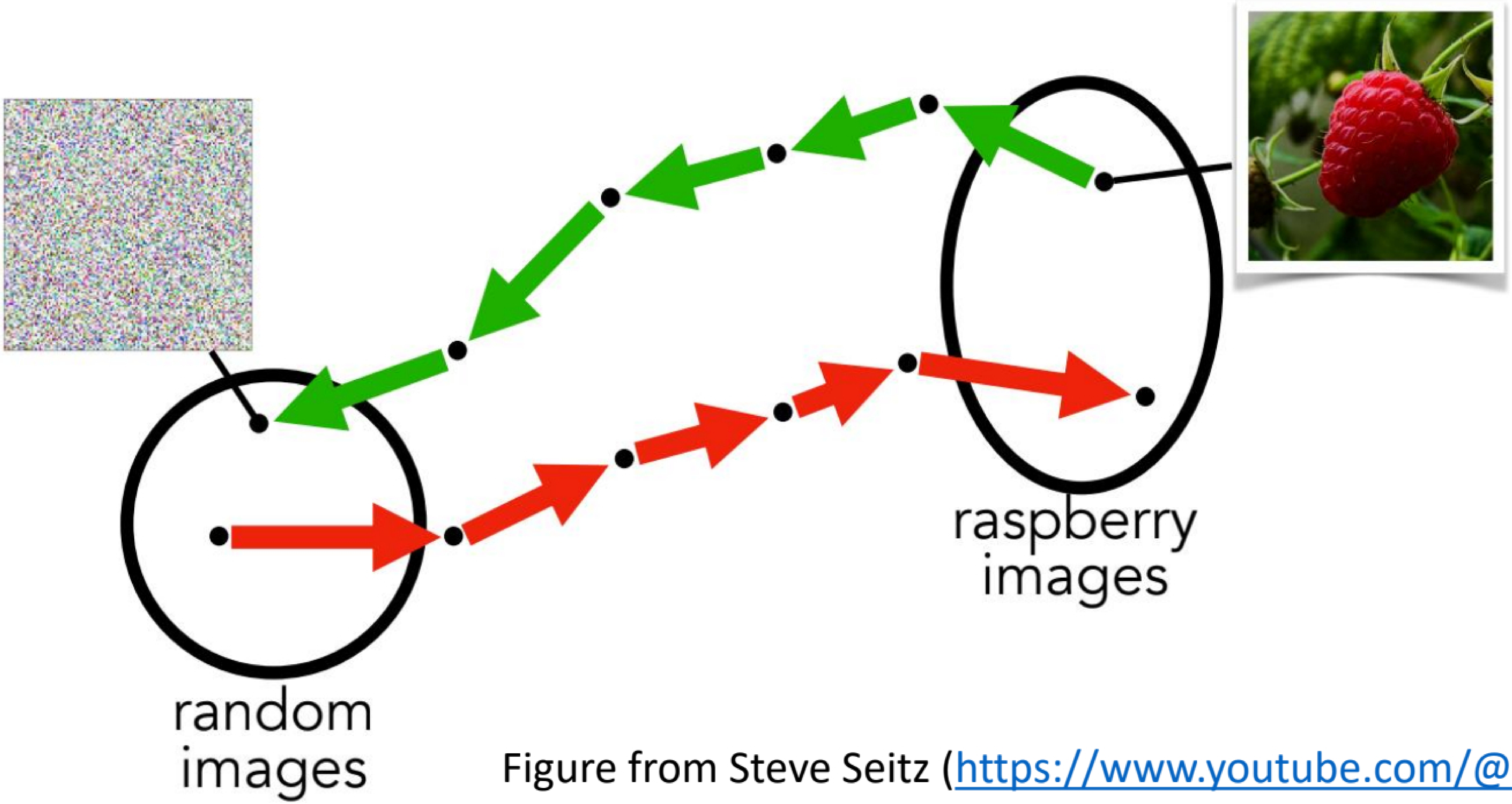
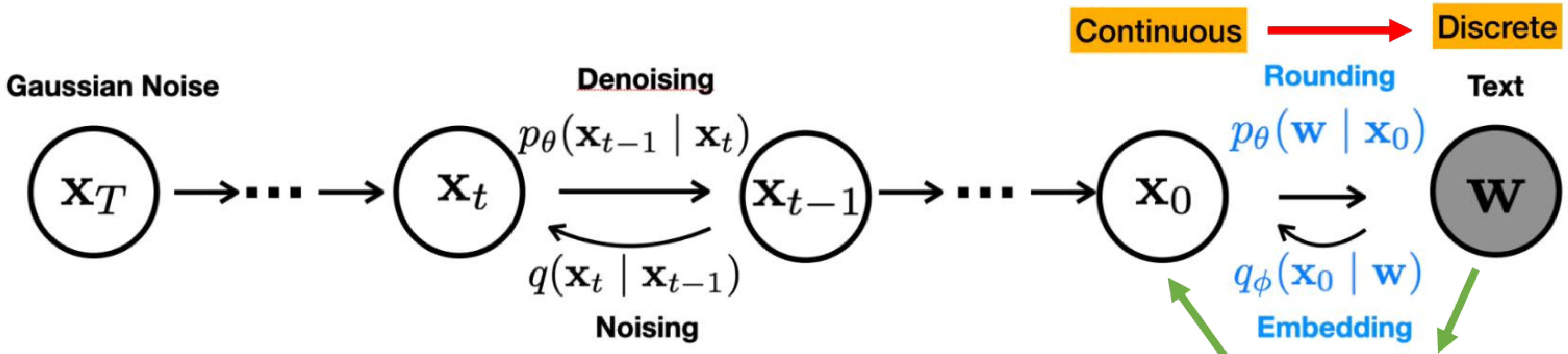


Figure from Steve Seitz (https://www.youtube.com/@g5min)

Diffusion in text: DiffusionLM



Rounding: $\hat{w}_i = \underset{w_i}{\operatorname{argmax}}\ p_\theta(w_i \mid x_0[i])$

Might get $w_1 \neq \overrightarrow{\text{cat}}$

$$\overrightarrow{\text{cat}} = \begin{bmatrix} -3 \\ 1.2 \\ -6.4 \\ \dots \\ -1.4 \\ 7.1 \end{bmatrix}$$

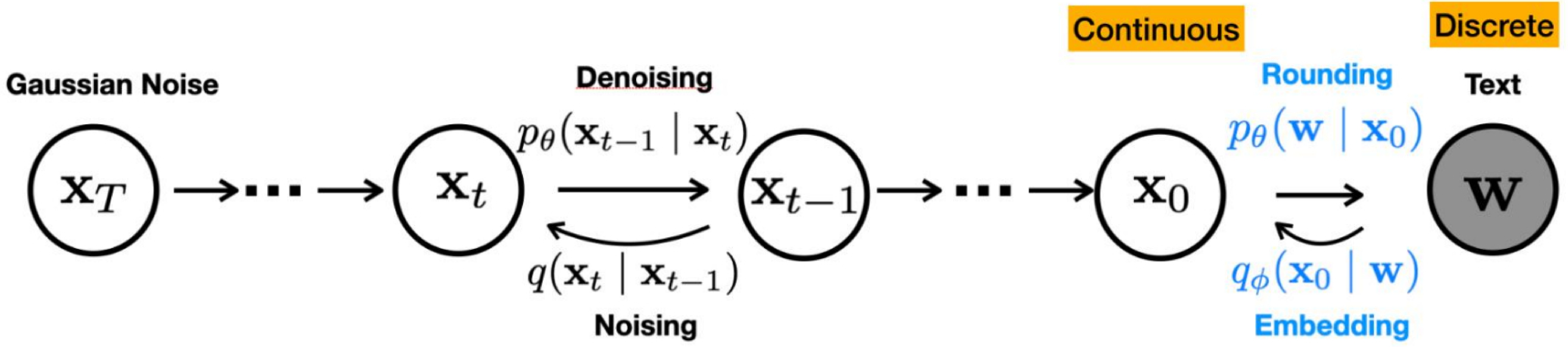
maximize probability of Embedding

$$\mathcal{L}_{\text{e2e}} = \underset{x_0 \sim q_\phi}{\mathbb{E}} \left[\boxed{\mathcal{L}_{\text{simple}}(x_0)} - \boxed{\log p_\theta(\mathbf{w} \mid x_0)} \right]$$

reconstruction loss

Figure from Xiang Lisa Li (https://web.stanford.edu/class/cs224u/slides/lisa-224u-diffusion.pdf)

Diffusion in text: DiffusionLM



maximize probability of Embedding

$$\mathcal{L}_{\text{e2e}} = \mathop{\mathbb{E}}_{x_0 \sim q_\phi} \left[\boxed{\mathcal{L}_{\text{simple}}(x_0)} - \boxed{\log p_\theta(\mathbf{w} \mid x_0)} \right] \Rightarrow p_\theta(\mathbf{w} \mid \mathbf{x}_0) = \prod_{i=1}^{n} p_\theta(w_i \mid x_i)$$

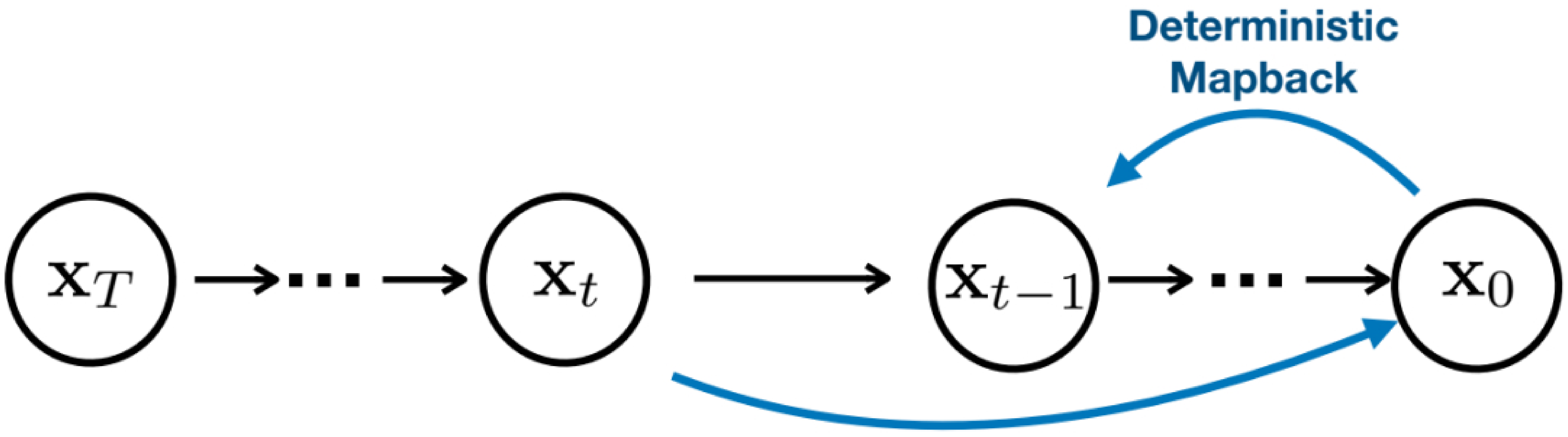
reconstruction loss

$$\mathcal{L}_{\text{simple}}(x_0) = \sum_{t=1}^{T} \mathop{\mathbb{E}}_{q(\mathbf{x}_t \mid \mathbf{x}_0)} ||\hat{\mu_\theta}(\mathbf{x}_t, t) - \mu_{t-1}(\mathbf{x}_t, \mathbf{x}_0)||^2$$

Can we reduce rounding error during training?

Figure from Xiang Lisa Li (https://web.stanford.edu/class/cs224u/slides/lisa-224u-diffusion.pdf)

Diffusion in text: DiffusionLM



$$\mathcal{L}_{\text{simple}}(x_0) = \sum_{t=1}^{T} \mathop{\mathbb{E}}_{q(x_t|x_0)} ||\hat{f}_\theta(x_t, t) - x_0||^2$$

Intuition: Training to predict x_0 at each diffusion step makes predicted x_0 more precise and reduce rounding error.

Figure from Xiang Lisa Li (https://web.stanford.edu/class/cs224u/slides/lisa-224u-diffusion.pdf)

Latent diffusion for text generation

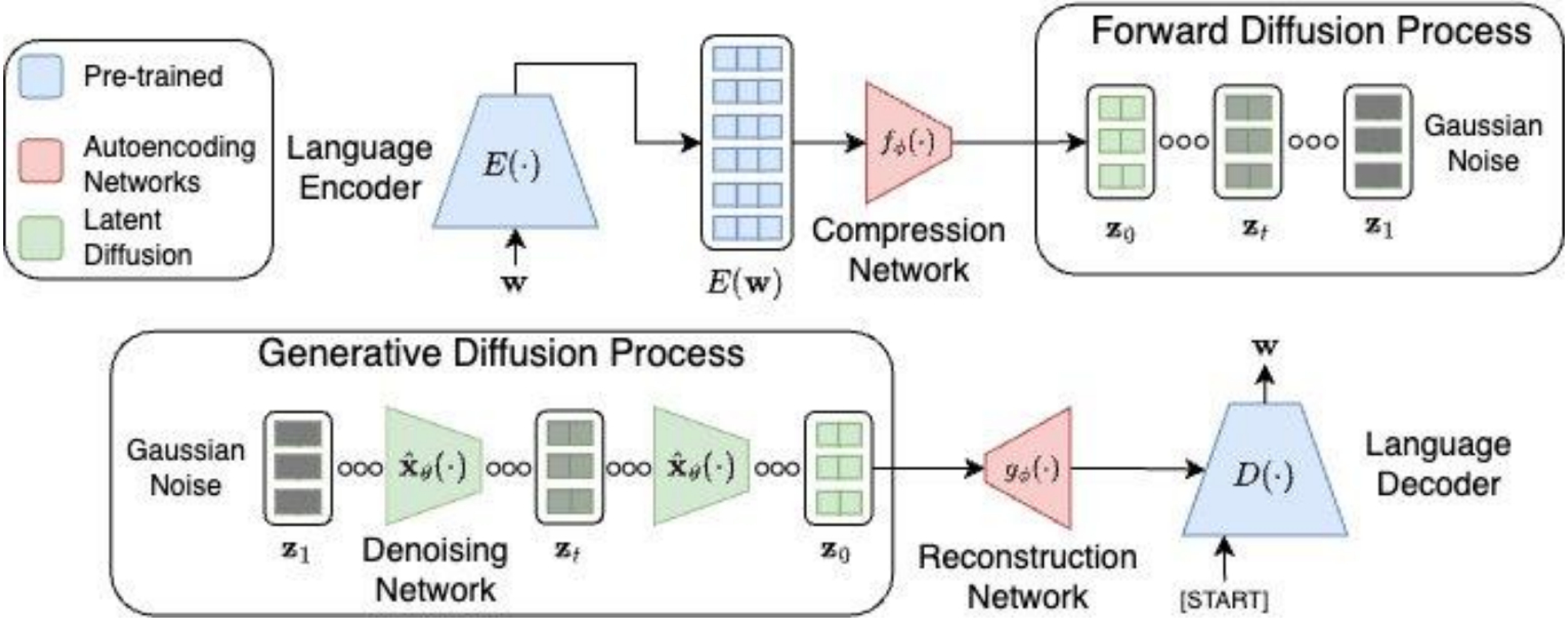


Figure from https://arxiv.org/abs/2212.09462

Semantic drift instead of rounding error!

Latent diffusion for text generation

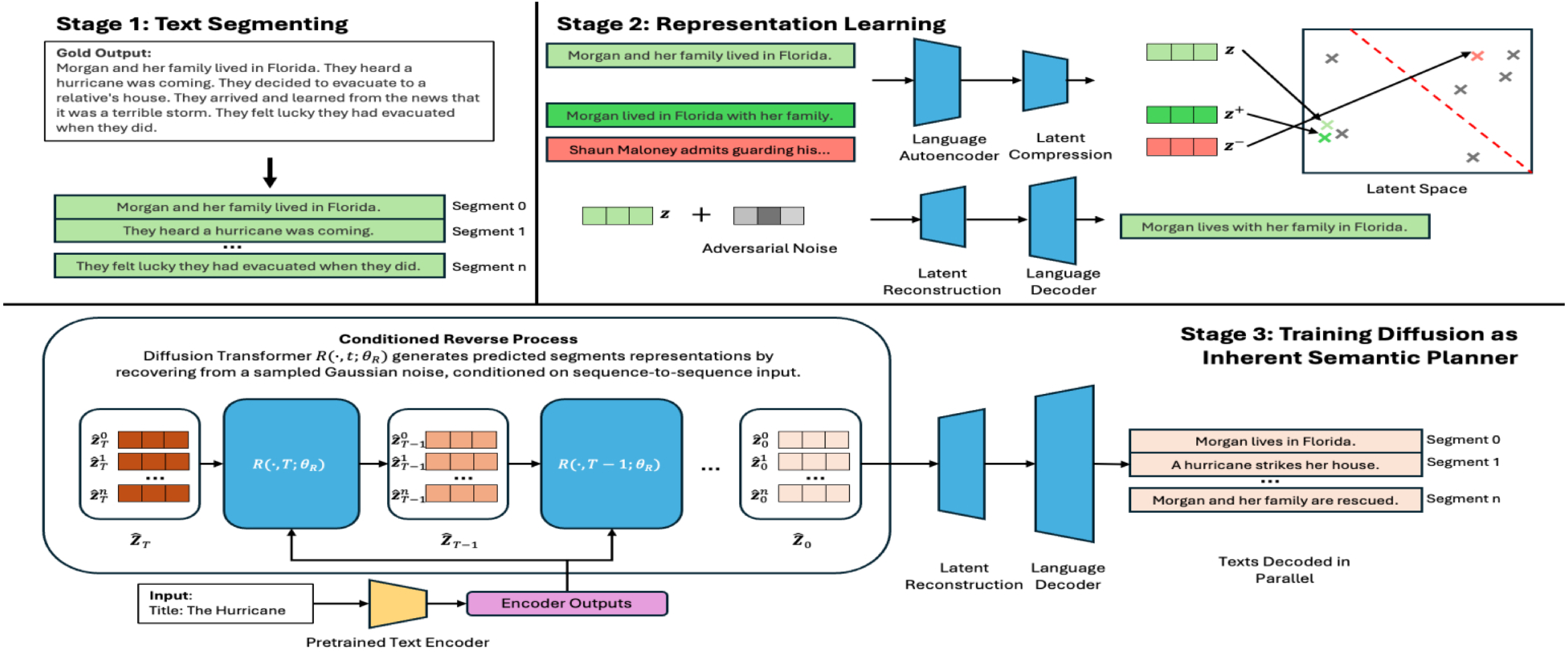
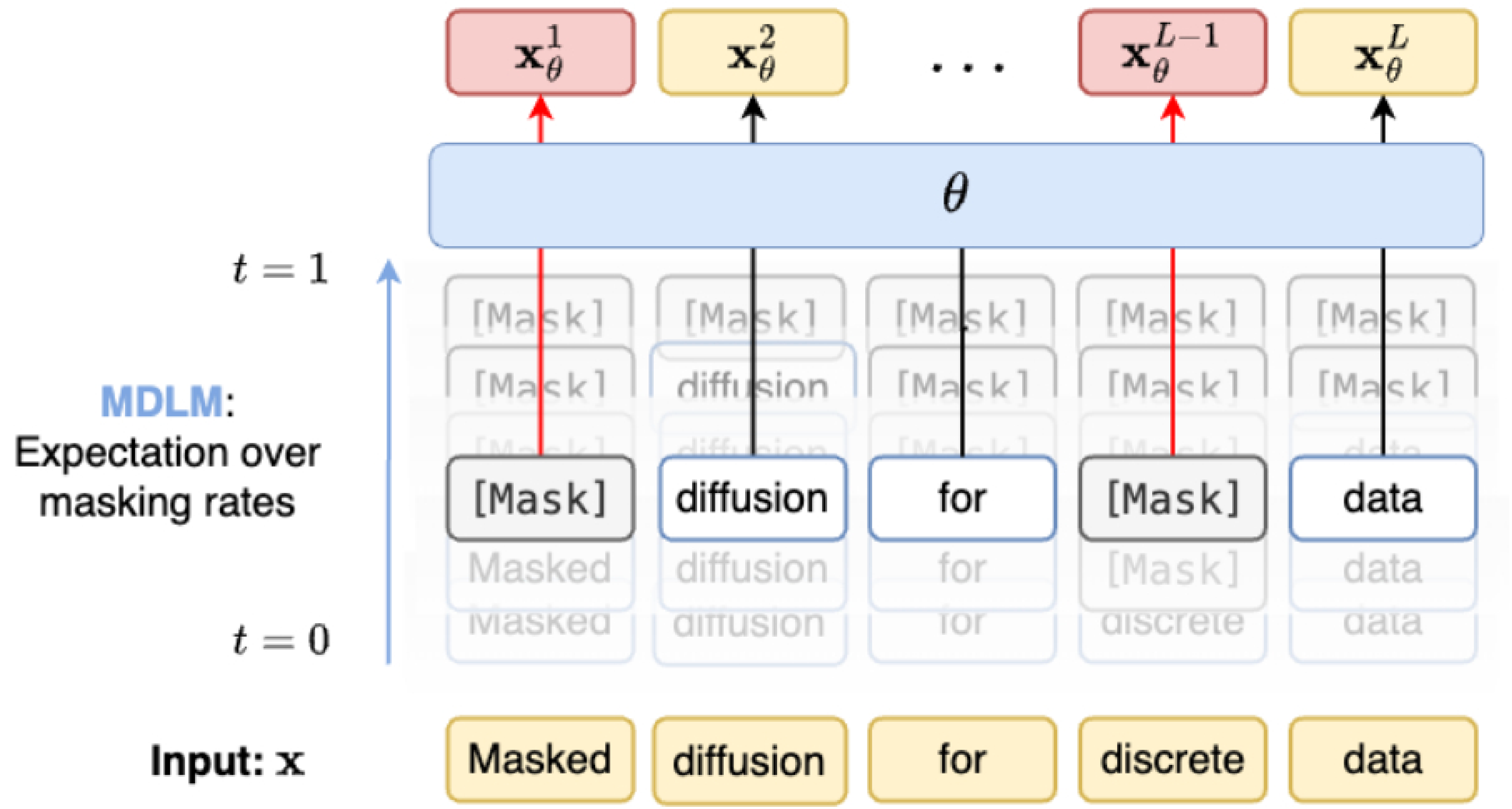


Figure from https://arxiv.org/pdf/2412.11333

Masked Diffusion LMs (NeurIPS 2026)

Diffusion Training: Average of unmasking losses

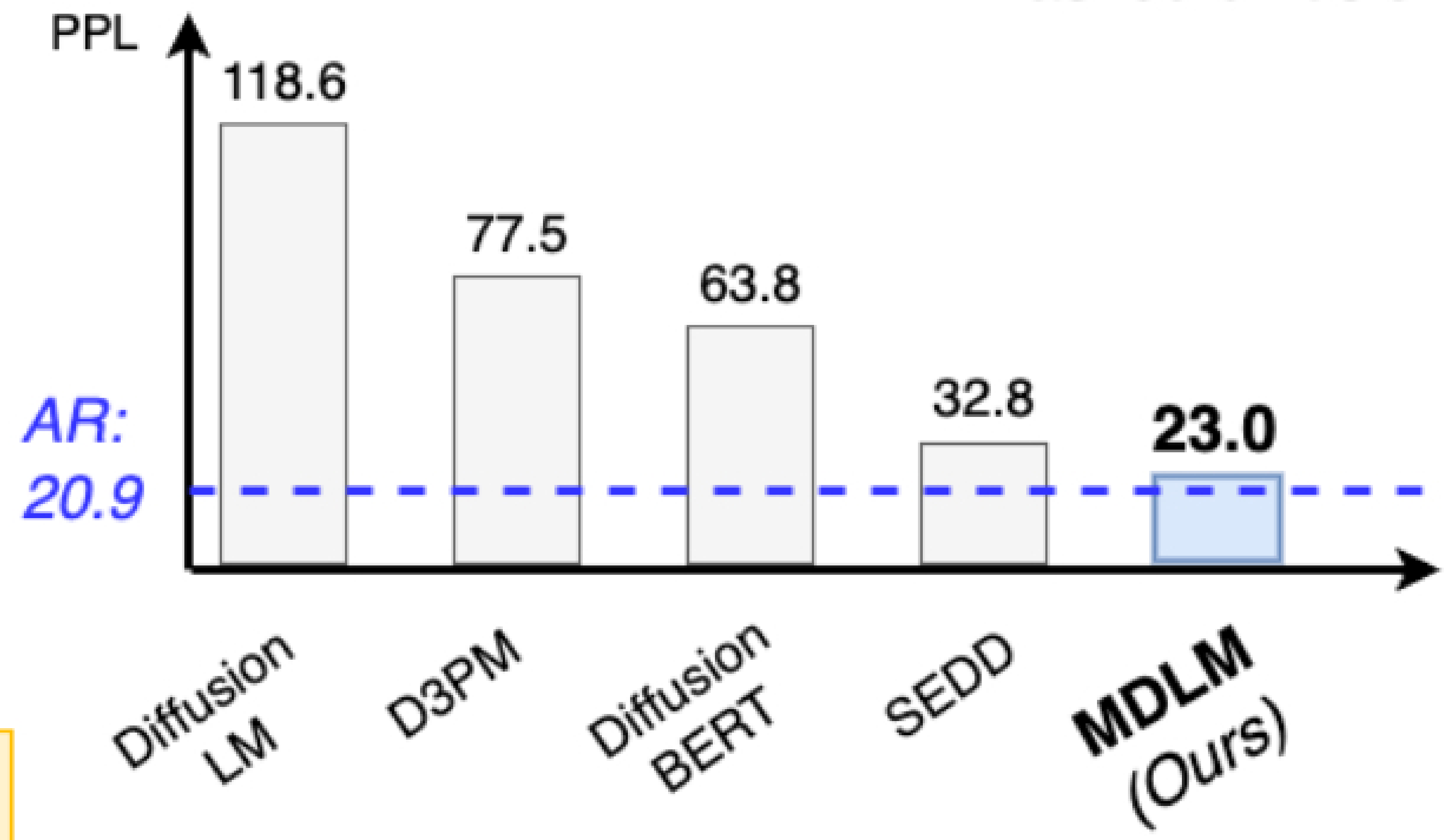


Randomly replace tokens with [MASK] or another token.

Figure from https://neurips.cc/virtual/2024/poster/95622

Simplified Masked Diffusion LM

- Masking rate is random, not fixed
- Objective is a variational lower bound
- Admits fast ancestral sampling
- Objective is a simple average of MLM losses
- ✓ Improved implementation relative to previous masked diffusion



Perplexity scores (lower is better).

Text-in-Image Diffusion: DeepFloyd

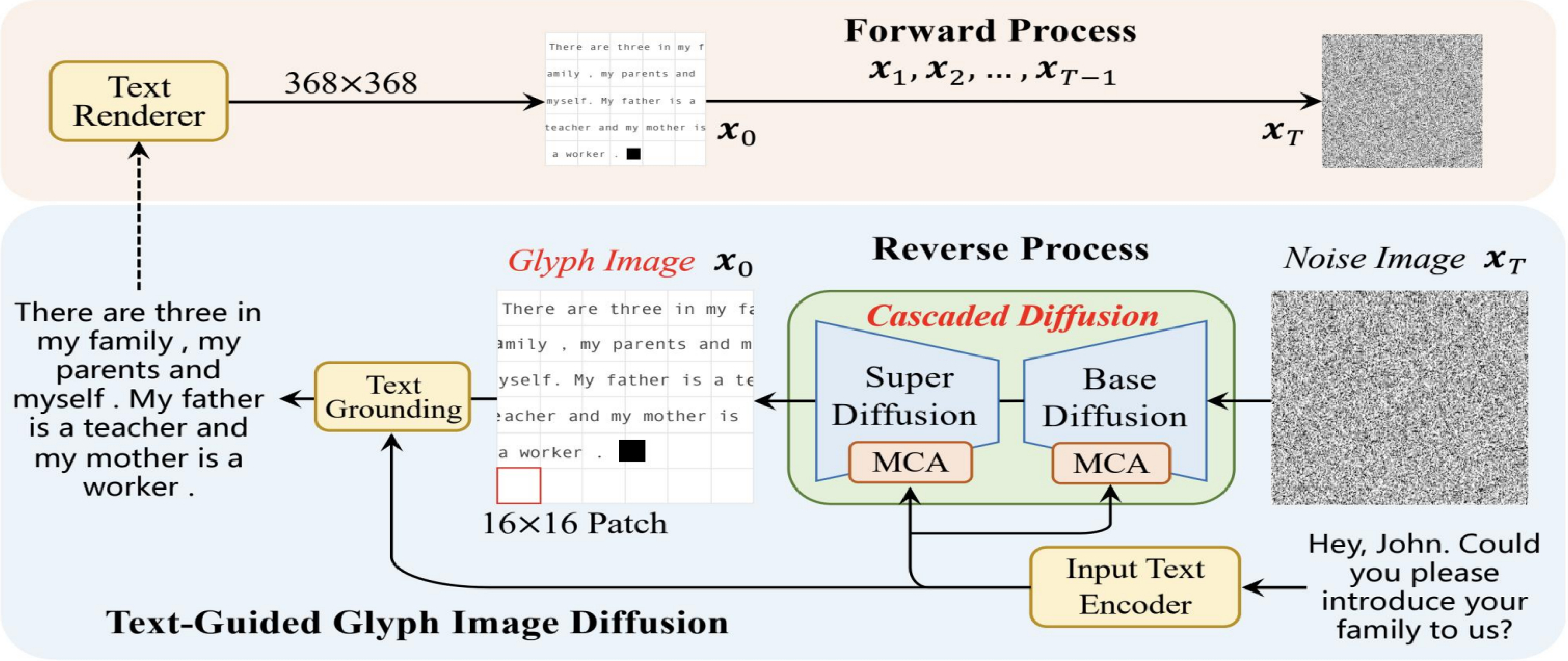


Figure from https://arxiv.org/pdf/2412.11333

Thank you! 😁